

Jahangirnagar University
Institute of Information Technology
Course Code: ICT- 2204
Course Title: Operating Systems Lab
Lab Quiz - 1

Total Marks: 10 Set-A Date: 17/06/2026 Time: 15 mins.

No.	Question	Marks
1	Create a file named student_A.txt in your home directory.	1
2	List all files (including hidden) in long format, sorted by modification time.	1
3	Create three files a1.txt, a2.txt, and b1.txt in a single command.	2
4	Use a single command with a metacharacter to list only files starting with a.	2
5	Create directory lab1, then create subdirectories week1 and week2 inside it in a single command.	2
6	Move into week1 and display your current path.	2
7	Set permissions of student_A.txt to: owner = read/write, group = read, others = none. Verify with ls.	2
8	Display the first 5 and the last 5 lines of notes.txt.	2
9	Search logfile.txt for the word error ignoring case and show line numbers.	1

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Total Marks: 10 Set-B Date: 17/06/2026 Time: 15 mins.

No.	Question	Marks
1	Create a directory school and, in a single command, create files report.txt, notes.txt, and .secret.txt inside it.	1
2	Open report.txt in vim, add three lines of text, then save and quit.	1
3	Copy report.txt to school/lab1 and rename the copy as report_backup.txt.	2
4	Count the number of lines, words, and characters in notes.txt.	2
5	Recursively copy directory school to school_copy in a single command.	2
6	Rename the directory school_copy to school_backup.	2
7	Set script.sh to permissions rwxr-xr-x using absolute mode and verify with ls.	2
8	Show all lines of notes.txt that do not contain the word debug.	2
9	Delete the file b1.txt from your current directory.	1

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Total Marks: 10 Set-C Date: 17/06/2026 Time: 15 mins.

No.	Question	Marks
1	Create directory project, then inside it create files main.c, main.h, and a hidden file .config in a single command.	1
2	List contents of project in long format, including hidden files.	1
3	Using a single command with a metacharacter, list only files in project that end with .c.	2
4	Create main.c with touch, write two lines into it using cat, then append one more line using echo.	2
5	Display the contents of main.c with line numbers using cat.	2
6	Recursively search for the string TODO under the project directory.	2
7	Set main.c permissions to owner = read/write, group = read, others = read, using symbolic mode. Verify with ls.	2
8	Force remove the directory project/week1 and all its contents.	2
9	Remove the empty directory project/empty_dir.	1

Jahangirnagar University
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Course Code: ICT- 3102
Course Title: Operating Systems Lab
Lab Quiz - 2

Total Marks: 10 Set-A Date: 22/07/2026 Time: 15 mins.

A university computer laboratory receives printing requests from students throughout the day. Each print request is treated as a process and is executed in the order it arrives. Write a C program to simulate the FCFS CPU Scheduling algorithm. Your program should:

No.	Question	Marks
1	Accept the Arrival Time (AT) and Burst Time (BT) of each process.	2
2	Display the Gantt Chart.	3
3	Calculate Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT).	3
4	Display the average WT and TAT.	2

Sample Input

Enter number of processes: 4

Process	Arrival Time	Burst Time
P1	0	5
P2	2	3
P3	4	2
P4	6	4

Sample Output

Gantt Chart

| P1 | P2 | P3 | P4 |

0----5----8---10----14

Process	AT	BT	CT	TAT	WT
P1	0	5	5	5	0
P2	2	3	8	6	3
P3	4	2	10	6	4
P4	6	4	14	8	4

Average Waiting Time = 2.75

Average Turnaround Time = 6.25

Jahangirnagar University
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Course Code: ICT- 3102
Course Title: Operating Systems Lab
Lab Final Exam

Total Marks: 30 Set-B Date: 23/11/2025 Time: 1 hour.

No.	Question	Marks
1	Create a user named "Lab Final Exam-Set-B".	2
2	Assign a password to ensure security.	2
3	Show all the files in the user.	1
4	Create a shell script file named "Problem-2.sh"	1
5	Change the ownership of the file to the user "Lab Final Exam-Set-A".	2
6	Ensure it has all the necessary permissions for working with the file.	2
7	Solve the following problem using the given "Problem-2.sh".	8

A robot must deliver packages over N days.

Each day has a "load value":

If load > 50 → robot must use High-Fuel mode, costing 5 units

If $20 < \text{load} \leq 50$ → Medium-Fuel mode, costing 3 units

If load ≤ 20 → Eco-mode, costing 1 unit

Special condition:

If the robot uses High-Fuel mode for 2 consecutive days, the 3rd day must also use High-Fuel mode regardless of load, costing 5 units (forced mode).

This forced mode does not reset the counter.

Input Format

- First line: integer T
- For each test case:
 - o First line: integer N
 - o Second line: N space-separated integers (load values)

Output Format

For each test case, print the total fuel cost.

Constraints

$1 \leq T \leq 100$; $1 \leq N \leq 10^5$; $0 \leq \text{load} \leq 100$; Sum of all N across test cases $\leq 10^5$

Sample Input

```
1
4
55 60 10 50
```

Output: 18

No.	Question	Marks
8	Solve the following Round Robin with Context Switch Penalty (quantum = 4).	12

Process	Arrival Time	CPU Burst
P1	0	9
P2	1	4
P3	2	2
P4	3	8
P5	5	3

Conditions:

Use Round Robin

Every context switch costs 2 unit time, during which no process executes

CPU is idle if no process is available

Tasks:

Construct the Gantt chart including context-switch delays

Calculate:

Waiting time for each process

Turnaround time for each process

Average waiting and turnaround time

How many total units of time were wasted due to context switching?